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Chapter 1

class notes

1.1 Coulomb's Law

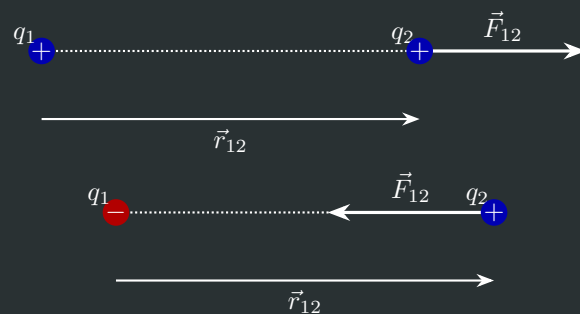
Coulomb's Law

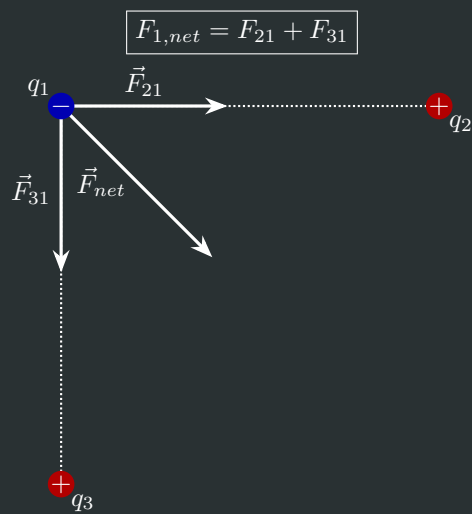
$$\vec{F}_{12} = k \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12} \quad (1.1)$$

Gives a force along the line of two charges.

If q_1 and q_2 have the same sign (i.e. charge) then the force must be repulsive. But if they have the *same* sign, then it must be attractive.

Because of Newton's second law, $F_{12} = -F_{21}$





Superpostion Principle

$$\vec{F}_{net} = \sum_i \vec{F}_i \quad (1.2)$$

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Electron Charge

$$e = -1.6 \times 10^{-19} C$$

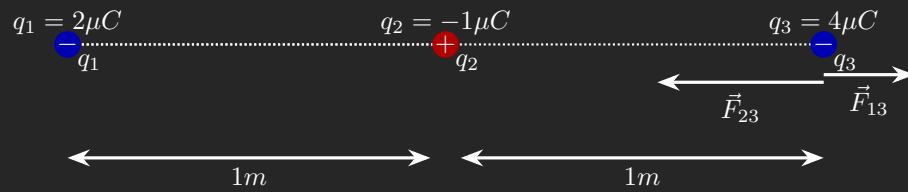
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k

$$k = 9 \times 10^9 \frac{NM^2}{C^2}$$

1.1.1 Examples

Example 1.1.1 $F_{3,net} = ???$



Using Coulomb's Law 1.1...

$$\vec{F}_{3,net} = \vec{F}_{31} + \vec{F}_{32}$$

$$\vec{F}_{31} = k \frac{q_3 q_1}{r_{31}^2} \hat{r}_{31}$$

$$\vec{F}_{32} = k \frac{q_3 q_2}{r_{32}^2} \hat{r}_{32}$$

$$k = 9 \times 10^9 \frac{NM^2}{C^2}, \quad q_1 = 2\mu C, \quad q_2 = -1\mu C, \quad q_3 = 4\mu C$$

$$\vec{F}_{31} = 1.800 \times 10^{-2} \text{ N}, \quad \vec{F}_{32} = -3.600 \times 10^{-2} \text{ N}$$

$$\vec{F}_{3,net} = -1.800 \times 10^{-2} \text{ N}$$

1.2 Electric flux and Field Lines

1.2.1 Field Lines

- Measure of electric field lines over an area

Electric Field

$$E = k \frac{q}{r^2} \quad (1.3)$$

Measure of the energy in an electric field at a point

Density of Field Lines

$$D = \frac{N}{4\pi r^2} \quad (1.4)$$

used when plotting

Number of Graphed Field Lines

$$N \equiv \frac{q}{\epsilon_0}, \quad E = \frac{N}{A} \quad (1.5)$$

1.2.2 Flux

Number of field lines passing through a surface

$$\Phi \propto N$$

For a point charge aligned at the center of a sphere it can just be defined as

$$\Phi \equiv EA$$

To calculate total over surface ...

$$\Phi \equiv \int_s \vec{E} d\vec{A}$$

1.3 5: potential energy

Guass's Law

$$\int E dS = \frac{Q_{enclosed}}{\epsilon_0} \quad (1.6)$$

Change in potential in const E field

$$\Delta U = q\vec{E}\ell \quad (1.7)$$

Change in potential in non static field

$$\Delta U = Q \int_a^b \vec{E} dr \quad (1.8)$$

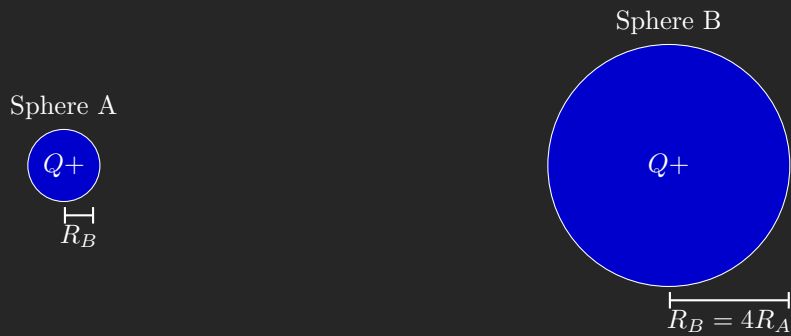
1.7 Conductors and Capacitance

1.7.1 Conductors

- Every point in a conductor is at equipotential, this is because the electric field at any point is zero.
- Continuing with Gauss's law, no matter the location of a charge contained *inside* a shell (but not overlapping the shell itself), the outside of the shell will have the same charge distribution, however the inside may not.
- When looking at the inverse, (a particle placed *outside* of a conductor), we find that the charges move to the surface to cancel out the exerted field, meaning that the inside of the object will have a net charge of 0.
- Importantly, this still applies when a cavity is made *within the conductor*
- This allows for the shielding of an object if it is placed within a conductor ¹

¹Think Faraday cage (maybe)

Example 1.7.1 Equipotential difference in two Spheres



$$V_A = - \int_{\infty}^{R_A} \vec{E}_A \cdot d\vec{l}, \quad V_B = - \int_{\infty}^{R_B} \vec{E}_B \cdot d\vec{l}$$

Given a point charge ...

$$\vec{E} = k \frac{Q}{r^2} \rightarrow - \int_{\infty}^r k \frac{Q}{r^2} = k \frac{Q}{r} \rightarrow$$

For a point charge...

$$V = k \frac{Q}{r}$$

$$V_A = k \frac{Q}{R_A}, \quad V_B = k \frac{Q}{4R_B}$$

$$4 \times V_A = V_B$$

Once connected, become a single conductor which wishes to reach equipotential. To find final charges ...

$$V_A = V_B \Rightarrow k \frac{Q_A}{R_A} = k \frac{Q_B}{R_B}$$

$$k \frac{Q_A}{R_A} = k \frac{Q_B}{4R_A}$$

$$4Q_A = Q_B$$

1.7.2 Capacitance

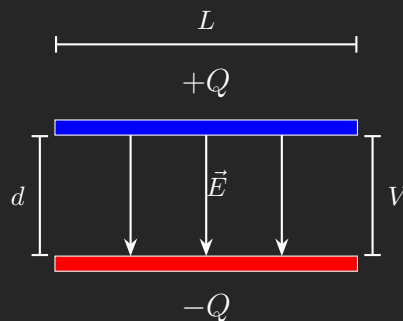
Capacitance

$$C \equiv \frac{Q}{\Delta V} \quad (1.9)$$

Units of Farads, or Coulombs per volt

- Measure of field between to oppositely charged objects
- Ex. A plate of Q and $-Q$ placed a distance d away from one another
- Stores the energy *in* the magnetic field induced between said objects

Example 1.7.2 Capacitance between two plates



$$d \ll L$$

To calculate E field between plates ...

$$E_{bot} = \frac{1}{2} \frac{\sigma}{\epsilon_0}, \quad E_{top} = \frac{1}{2} \frac{\sigma}{\epsilon_0}$$

$$\sigma = \frac{Q}{A}$$

$$E_{bot} = \frac{1}{2} \frac{Q}{\epsilon_0 A}, \quad E_{top} = \frac{1}{2} \frac{Q}{\epsilon_0 A}$$

Since field lines are the same direction...

$$E_{tot} = \frac{Q}{\epsilon_0 A}$$

To find capacitance...

$$C \equiv \frac{Q}{\Delta V}$$

$$|\Delta V| = \int_{bot}^{top} \vec{E} \cdot d\vec{l} \Rightarrow \Delta V = \frac{Q}{\epsilon_0 A} \int_{bot}^{top} dl$$

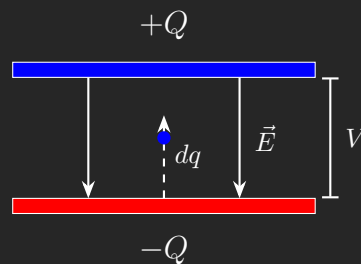
$$\Delta V = \frac{Q}{\epsilon_0 A} d$$

$$C = \frac{\epsilon_0 A}{d}$$

Capacitance for Parallel Plates

$$C = \frac{\epsilon_0 A}{d} \quad (1.10)$$

Example 1.7.3 Parallel Plate Capacitor



To calculate the work needed to move a particle from the bottom plate to the top plate

$$dW = dqEd$$

Since we know that the energy from the electric field is equal to $\frac{V}{d} \dots$

$$dW = dq \frac{V}{d} d$$

$$dU = Vdq$$

This holds true for *any system*. Note that dU is *not* constant because for each particle moved, it becomes requires more work for the next.

To calculate potential of a capacitor. . .

$$U = \int_0^Q V dq \Rightarrow U = \int_0^Q \frac{q}{C} dq$$

Thus . . .

$$U = \frac{1}{2} \frac{Q^2}{C}$$

Potential of a Capacitor

$$U = \frac{1}{2} \frac{Q^2}{C} \quad U = \frac{1}{2} CV^2 \quad U = \frac{1}{2} QV \quad (1.11)$$

Energy density of a capacitor

$$u = \frac{1}{2} \epsilon_0 E^2 \quad (1.12)$$

1.8 Capacitors

What happens if we pull apart a parallel plate capacitor?

- Q remains constant
- E increases
- ΔV increases
- C decreases
- U increases

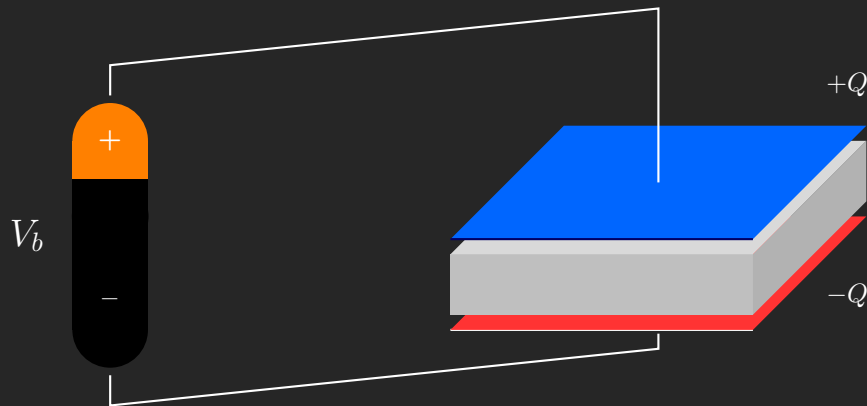
1.8.1 Dielectrics

- Material where positive and negative charges in individual molecules can change, but not through the entire material
- When placed between a capacitor all of the charges align themselves to the field
- This creates a material full of little dipoles
- This induced field is the *opposite* of the one from the plates
- This has the effect of *reducing* the electric field within the capacitor, which then reduces the potential diff (V), which then *increases* the capacitance

$$C = \frac{Q}{V}$$

- The amount a capacitance increases is called the **Dielectric Constant** and is calculated via $C_{old} = \kappa C_{new}$

Example 1.8.1 Adding a Dielectric to a Capacitor



- Capacitance increases
- $V_c = V_b$
- Because $C = \frac{Q}{V} \Rightarrow Q \uparrow$
- Because $U = \frac{1}{2} Q V_c \Rightarrow U \uparrow$

1.8.2 Capacitors in Parallel

- When wired in parallel they can be treated as one capacitor with a larger area
- this equivalent capacitor has...

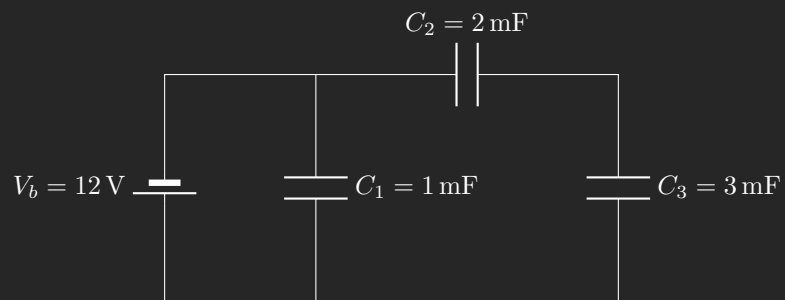
$$V_{equiv} = V_1 = V_2 \quad C_{equiv} = C_1 + C_2, \quad Q_{equiv} = Q_1 + Q_2$$

1.8.3 Capacitor in series

- When wired in series they act like a capacitor with a large distance
- This equivalent capacitor has ...

$$Q_{equiv} = Q_1 = Q_2 \quad \frac{1}{C_{equiv}} = \frac{1}{C_1} + \frac{1}{C_2}, \quad V_{equiv} = V_1 + V_2$$

Example 1.8.2 Combination of Capacitors



Simply into single capacitors

$$C_2, C_3 \rightarrow \frac{1}{C_{23}} = \frac{1}{C_2} + \frac{1}{C_3} = 1.2 \text{ mF}$$

$$C_1, C_{23} \rightarrow C_1 + C_{23} = C_{123} = 2.2 \text{ mF}$$

1.9 Kirchhoff's Rules

Kirchhoff Voltage Rule

$$\sum \Delta V_n = 0 \quad (1.13)$$

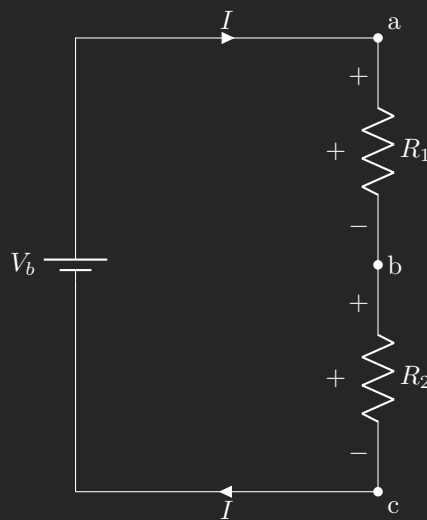
In a **closed** circuit, if you return to the same location, no matter what path you take, the change in voltage will be 0. Very similar to conservation of energy in a magnetic field.

Kirchhoff Current Rule

$$\boxed{\sum I_{in} = \sum I_{out}}, \text{ at any node} \quad (1.14)$$

At any node (location where two or more paths/wires meet), the current in must be equal to the current out (of said node)

Example 1.9.1 Single Loop Example



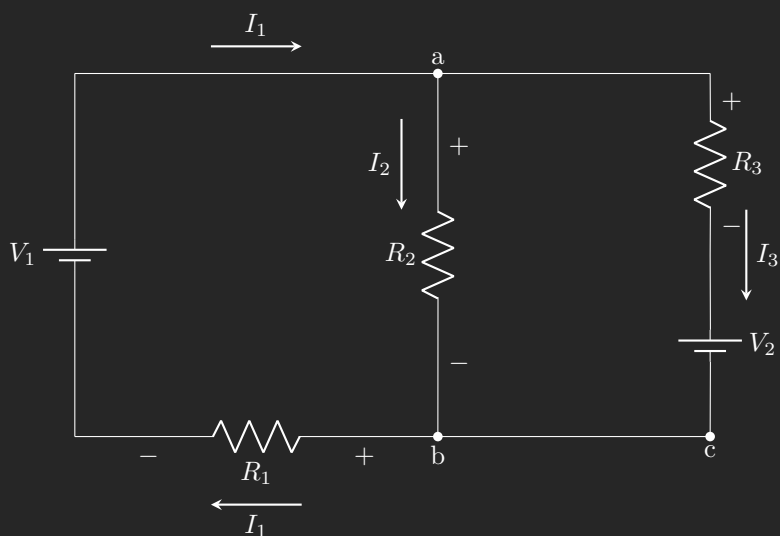
Can be simply solved by using Ohms law and doing $I = \frac{V_b}{R_1 + R_2}$, or using Kirchhoff

Using Kirchhoff's Voltage Rule (1.13) we find the drop in current between
 $a \rightarrow b \rightarrow c \rightarrow V_b$

$$\Delta V_1 = +IR_1, \quad \Delta V_2 = +IR_2, \quad \Delta V_3 = -V_b$$

$$IR_1 + IR_2 - V_b = 0 \quad \Rightarrow \quad I = \frac{V_b}{R_1 + R_2}$$

Example 1.9.2 Two Loop Example



Using Kirchhoff's Current Rule (1.14) we find that $I_1 + I_2 + I_3 = 0$. To solve this we need equations with I_1, I_2, I_3

Using Kirchhoff's Voltage Rule (1.13) on the *outer* loop we get

$$I_3 R_3 + V_2 + I_3 R_1 - V_1$$

On the inner/left loop

$$I_2 R_2 + I_1 R_1 - V_1$$

1.12 Magnetism

1.12.1 Bar Magnets

- Has a north and south pole
- like sides repel opposite sides attract
- the field *looks* like a dipole setup when measuring the field, however it is not
- if the magnet is split in half it becomes two more bar magnets

Magnetic Force

$$\vec{F} = q\vec{v} \times \vec{B} \quad (1.15)$$

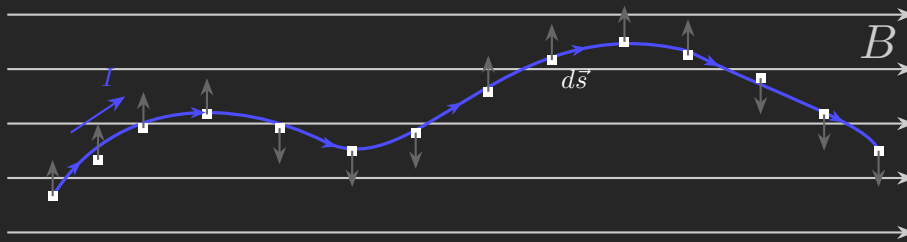
Movement of a particle in a magnetic field

1.13 Forces and Torques on Currents

Force on a Current Carrying Wire

$$\vec{F} = I\vec{L} \times \vec{B} \quad (1.16)$$

Example 1.13.1 Force on a Curved Wire Segment



- When looking at a curved wire inside a magnetic field, the only relevant part is the height difference (or perpendicular component). This is because the parallel component doesn't have an effect, as its cross product with the field is 0
- This means to find the total force, you sum up all the vertical components
- this is the same as adding all of them together
- which is the same thing after finding the length between the start and end point (denoted as L) and taking the cross product of it with the field
- This gives us the final equation of $F_{wire} = I\vec{L} \times \vec{B}$

Torque On Current Loop

$$\tau_{loop} = IAB \sin \theta \quad (1.17)$$

Magnetic Dipole Moment

$$\mu = NI\vec{A} \quad (1.18)$$

Where N is number of coils

Torque on a Loop

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad (1.19)$$

Torque Work

$$W = \int \tau d\theta \quad (1.20)$$

Potential Energy of Dipole in B Field

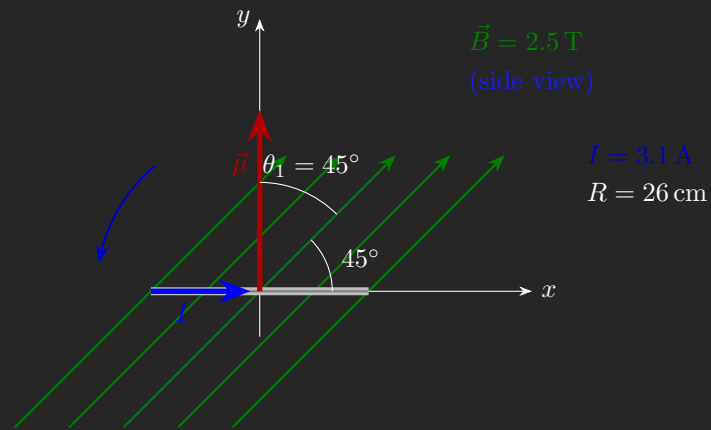
$$U(\theta) = -\vec{\mu} \cdot \vec{B} \quad (1.21)$$

Work on a Current Carrying Loop in a Magnetic Field

$$W = \int_{\theta_o}^{\theta_2} (-\mu B \sin \theta) d\theta \quad (1.22)$$

1.13.1 Examples

Example 1.13.2 Forces and Torques on Current



Use 1.21 and 1.18

$$U_{\theta_1} = -\mu B \cos 45^\circ$$

$$U_{\theta_2} = -\mu B \cos 135^\circ$$

$$\Delta U = U(\theta_2) - U(\theta_1)$$

$$\mu = NIA$$

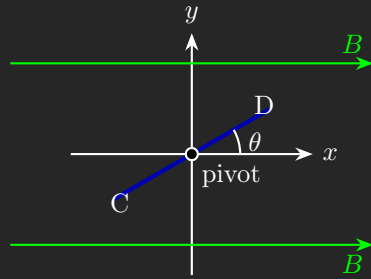
Example 1.13.3 Square Current Loop with Gravity

A uniform magnetic field $B = 1.8T$ points in the $+x$ direction. A square loop with side length $d = \text{cm}$, number of turns 20, and current per turn $I = .85A$ pivots without friction around a vertical pin aligned with the $-y$ -axis.

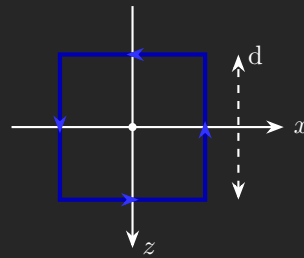
The left panel shows a side view of the loop in the xy -plane. The loop is tilted at an angle $\theta = 30^\circ$ relative to the vertical xz -plane. The pivot is at the origin. The segment of the loop labeled CD lies in a plane that is rotated about the z -axis by the angle θ . Gravity acts downward along the direction, and the magnetic field points horizontally along $+x$.

The right panel shows the top-down view of the loop, which is square with both horizontal and vertical sides of length d . Segment CD lies across the top edge of the square, and current circulates counterclockwise in the loop when viewed from above.

A mass M is hung from one vertical side of the loop, either from point C or point D. The system is in equilibrium at the angle θ described above.



Side View



Top-Down View

$$N = 12, \quad I = .85, \quad I_{eff} = 10.2, \quad d = .2\text{m}, \quad B = 1.8T, \quad \theta = 30^\circ, \quad g \approx 9.7\text{m/s}^2$$

Using RHR, the magnetic moment points in the direction of $+y$. Using 1.18...

$$\mu = NIA \Rightarrow NId^2$$

Using 1.19...

$$|\tau_m| = NId^2B \cos \theta$$

1.14 Biot-Savart Law

Biot-Savart Law

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{s} \times \hat{r}}{r^2} \quad (1.23)$$

Where $d\vec{s}$ is a length of the wire, and r is the distance from said length segment. $\hat{r}$ is a vector from the wire to the point. $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$. ds takes direction of current.

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μ_0

$$\mu_0 = 4\pi \times 10^{-7} \frac{\text{T} \cdot \text{m}}{\text{A}}$$

Biot-Savart Law for a infinite straight wire

$$B = \frac{\mu_0 I}{2\pi R} \quad (1.24)$$

where R is perpendicular distance

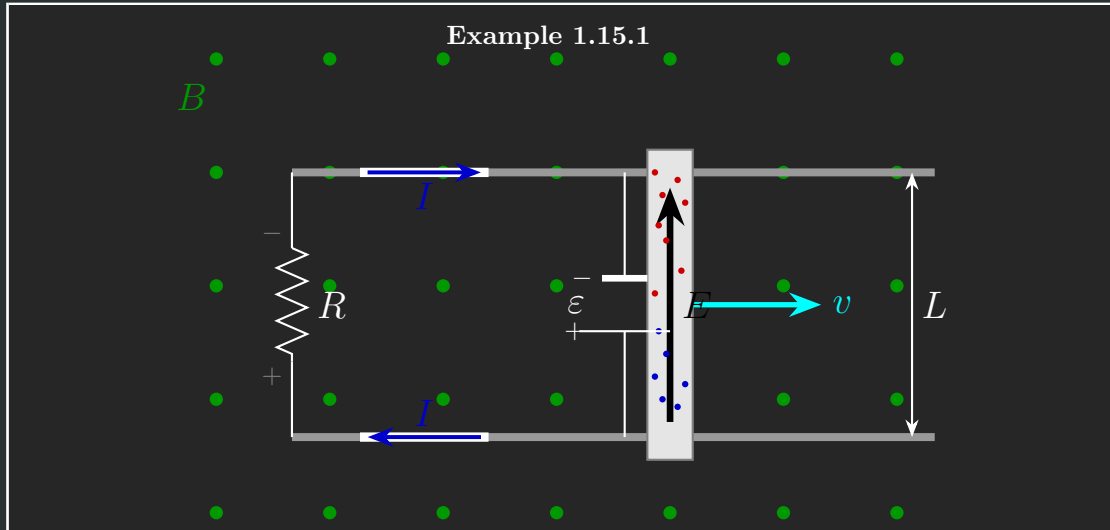
1.15 Motional EMF

Solving when the magnetic field is changing on the object.

If a bar is moving through a magnetic field, the proton and electrons will be pushed to *opposite* ends of the bar, creating an *electric* field.

This field allows for a circuit to then be attached, in which case current will flow through depending on the velocity of the bar.

There is a lorentz force pushing the bar in the *opposite* direction of travel.



E field of a bar moving through B field

$$E = vB \quad (1.25)$$

If this bar is attached to rails that connect to a circuit, it will drive current through said current.

Potential Difference of Bar in B field

$$\varepsilon = vBL \quad (1.26)$$

Motional EMF of a generator

$$\varepsilon = \omega AB \cos(\omega t) \quad (1.27)$$

Chapter 2

disc

2.1 pset1

2.1.1 a

It is possible that the third option c could be true. This is because we see an exponential acceleration in the Y direction, which correlates with W^2

2.1.2 b

$$a = F \quad t = \frac{L}{V_0} \rightarrow y_f = \frac{1}{2} \frac{FL^2}{V_0}$$

2.1.3 c

$$\frac{w}{2} = \frac{1}{2} \frac{F_{max} L^2}{mv_0^2} \quad F_{max} = \frac{mv_0^2 w}{L^2}$$

2.1.4 d

$$\tan(\theta) = \frac{v_y}{v_o} = \frac{w}{l}$$
$$v_y = \frac{F}{m} \times \frac{L}{v_0}$$

2.2

2.2.1

It has a net force of 0. This is because the x and y components cancel each other out.

2.2.2

The y component would cancel, as it is symmetric around the particle. X would not form a triangle to find r to use $u_g = \frac{-Gm_1m_2}{r^2}$

$$F_g = -\frac{GMm}{\left(\frac{\sqrt{5}a}{2}\right)^2}$$

$$F_{g,x} = -\frac{16Mm}{g\sqrt{5}a^2}$$

2.2.3

2.2.4

The physical argument is that the object is quite far away, and thus we can use a small angle aprox.

$$\sqrt{(x - a/2)^2 + (a/2)^2} \rightarrow \approx x$$

$$\frac{GMm}{x^2}$$

2.3 RL stuff lol

it is the fourth one

2.3.1

$$u_c = -\frac{4\sqrt{2}GMm}{a}$$

2.3.2

$$\Delta u = \frac{4\sqrt{2}GMm}{a} = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{8\sqrt{2}GM}{a}}$$

2.3.3

$$u_1, u_2 = -\frac{2GMm}{a}$$

$$u_3, u_4 = \frac{-2GMm}{\sqrt{5}a}$$

$$u = u_1 + u_2 \cdots = -\frac{4GMm}{a}\left(1 + \frac{1}{\sqrt{5}} = Ur\right)$$

2.3.4

no, the middle has *higher* potential energy

2.3.5

2.4

2.4.1

$$p^2 = \frac{4a^3\pi^2}{GM}$$

2.4.2

$$\omega = 2\pi/p \rightarrow \omega^2 = \frac{4\pi^2 GM}{\cancel{4a^3\pi^2}} = \frac{GM}{r^3}$$

2.4.3

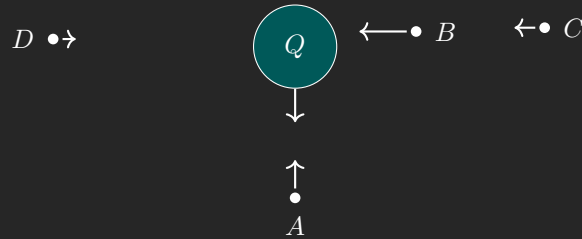
$$\begin{aligned} Dw^a R6a + b \\ a = \frac{D}{m} w^a R^{a+b} = \omega^2 R \\ \omega^{2-a} = \frac{D}{m} R^{a+b-1} \end{aligned}$$

2.4.4

2.5 w2

2.5.1 q1

Example 2.5.1 The configuration below shows some points around a positive charge Q . If a positive test charge q is placed at point A, at a distance r_A from Q , calculate the magnitude of the force between the charges in terms the given variables.



P1) Find magnitude

Because we're finding a charge diff, we use 1.1

$$\vec{F}_{12} = k \frac{Qq}{r_A^2}$$

P2) Force strength

Based on distance to the *field* of Q

$$F_B > F_A > F_C > F_D$$

P4) Electric Field Mag

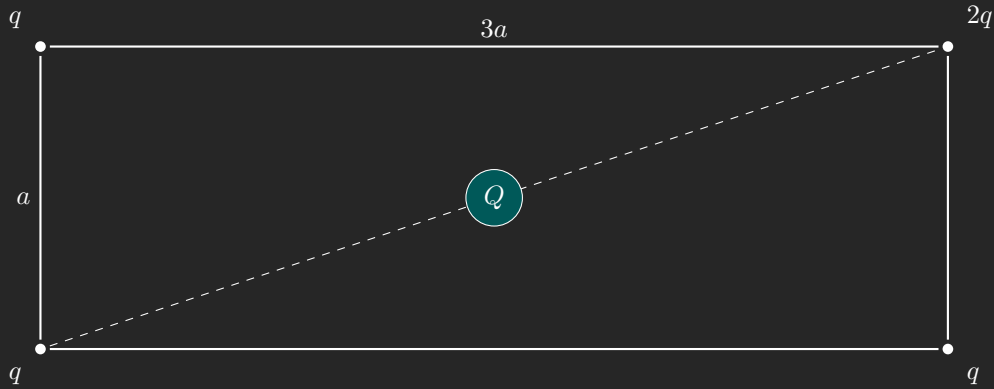
Because a test charge placed in a field exerts doesn't have any force of its own, we use a modified version of 1.3

$$E = k \frac{Q}{r_a^2}$$

Based on above...

$$2Q \propto 2E, \quad 2q \propto 2E$$

Example 2.5.2 Electric Field Due to Point Charges Four point charges $2q$, q , q , and q are placed at the corners of a rectangle of dimensions a and $3a$ as shown in the figure. A fifth charge Q is placed at the center of the rectangle. Our task is to compute the electric field at the center of the rectangle, and then determine the force on Q .



P1) Find the mag of field charges

$$r = \sqrt{\left(\frac{3a}{2}\right)^2 + \frac{1}{(2a)^2}}$$

$$r = \sqrt{\frac{9a^2}{4} + \frac{1}{4a^2}}$$

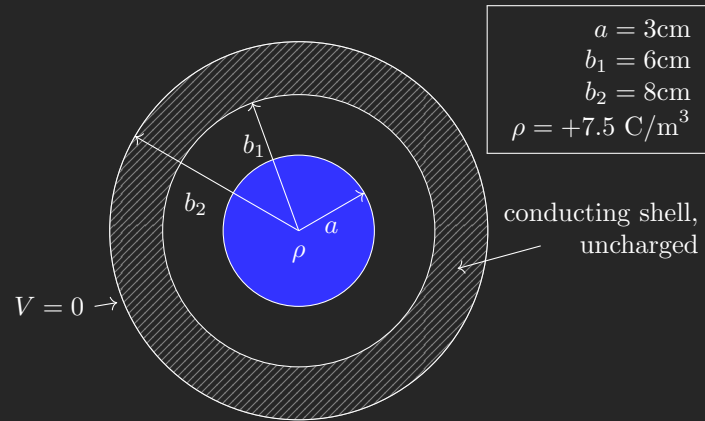
$$r = \sqrt{\left(\frac{10a^2}{4}\right)}$$

$$r = \frac{\sqrt{5}}{\sqrt{2}}a$$

$$q = E_1 = \frac{2ka}{5a^2}, \quad 2q = E_2 = \frac{4ka}{5a^2}$$

P3) Find X and Y components

Example 2.6.1 Electric Potential in a System with Cylindrical Symmetry



Find the potential at $r = 10\text{cm}$

Find the energy enclosed using Gauss's Law (1.6)

$$\int \vec{E} dS = \frac{Q_{enc}}{\epsilon_0}$$

$$\lambda = \rho * \pi * a^2$$

$$Q_{enc} = \rho \pi a^2 L$$

$$E(2\pi r) = \rho \pi a^2$$

$$E = \frac{\rho a^2}{2\epsilon_0 r}$$

$$V(r) = \int_r^{b_2} \frac{\rho a^2}{2\epsilon_0 r}$$

$$V(r) = \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b_2}{r}\right)$$

2.7 Equations

$$\vec{F}_{12} = k \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12} \quad (1.1) \text{ (Coulomb's Law)}$$

$$\vec{F}_{net} = \sum_i \vec{F}_i \quad (1.2) \text{ (Superposition Principle)}$$

$$E = k \frac{q}{r^2} \quad (1.3) \text{ (Electric Field)}$$

$$D = \frac{N}{4\pi r^2} \quad (1.4) \text{ (Density of Field Lines)}$$

$$N \equiv \frac{q}{\epsilon_0}, \quad E = \frac{N}{A} \quad (1.5) \text{ (Number of Graphed Field Lines)}$$

$$\int E dS = \frac{Q_{enclosed}}{\epsilon_0} \quad (1.6) \text{ (Guass's Law)}$$

$$\Delta U = q \vec{E} \ell \quad (1.7) \text{ (Change in potential in const E field)}$$

$$\Delta U = Q \int_a^b \vec{E} dr \quad (1.8) \text{ (Change in potential in non static field)}$$

$$C \equiv \frac{Q}{\Delta V} \quad (1.9) \text{ (Capacitance)}$$

$$C = \frac{\epsilon_0 A}{d} \quad (1.10) \text{ (Capacitance for Parrel Plates)}$$

$$U = \frac{1}{2} \frac{Q^2}{C} \quad U = \frac{1}{2} C V^2 \quad U = \frac{1}{2} Q V \quad (1.11) \text{ (Potential of a Capacitor)}$$

$$u = \frac{1}{2} \epsilon_0 E^2 \quad (1.12) \text{ (Energy density of a capacitor)}$$

$$\sum \Delta V_n = 0 \quad (1.13) \text{ (Kirchhoff Voltage Rule)}$$

$$\boxed{\sum I_{in} = \sum I_{out}}, \text{ at any node} \quad (1.14) \text{ (Kirchhoff Current Rule)}$$

$$\vec{F} = q \vec{v} \times \vec{B} \quad (1.15) \text{ (Magnetic Force)}$$

$$\vec{F} = I \vec{L} \times \vec{B} \quad (1.16) \text{ (Force on a Current Carrying Wire)}$$

$$\tau_{loop} = IAB \sin \theta \quad (1.17) \text{ (Torque On Current Loop)}$$

$$\mu = N I \vec{A} \quad (1.18) \text{ (Magnetic Dipole Moment)}$$

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad (1.19) \text{ (Torque on a Loop)}$$

$$W = \int \tau d\theta \quad (1.20) \text{ (Torque Work)}$$

$$U(\theta) = -\vec{\mu} \cdot \vec{B} \quad (1.21) \text{ (Potential Energy of Dipole in B Field)}$$

$$W = \int_{\theta_o}^{\theta_2} (-\mu B \sin \theta) d\theta \quad (1.22) \text{ (Work on a Current Carrying Loop)}$$

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{s} \times \hat{r}}{r^2} \quad (1.23) \text{ (Biot-Savart Law)}$$

$$B = \frac{\mu_0 I}{2\pi R} \quad (1.24) \text{ (Biot-Savart Law for a infinite straight wire)}$$

$$E = vB \quad (1.25) \text{ (E field of a bar moving through B field)}$$

$$\varepsilon = vBL \quad (1.26) \text{ (Potential Difference of Bar in B field)}$$

$$\varepsilon = \omega AB \cos(\omega t) \quad (1.27) \text{ (Motional EMF of a generator)}$$

2.8 Constants

$$e = -1.6 \times 10^{-19} C \quad (1) \text{ (Electron Charge)}$$

$$k = 9 \times 10^9 \frac{NM^2}{C^2} \quad (2) \text{ (k)}$$

$$\mu_0 = 4\pi \times 10^{-7} \frac{T \cdot m}{A} \quad (3) \text{ (\mu}_0\text{)}$$

Glossary

Dielectric Constant Denoted as κ

$$C_{new} = \kappa C_{old}$$

The amount which the capacitance of a capacitor increases when a dielectric is placed between the charges surfaces [13](#)